The Usadel free-energy solver: equations, literature anchoring, and implementation

Methods documentation for `usadel_free_energy.py`

July 16, 2026

This document states every equation used by the solver, gives its literature source, and *derives* every place where our form differs from the cited one. Section 5 gives the high-level code structure. Units: energies in Δ_{00} (the T=0 bulk gap of Nb), lengths in $\xi_0 = \sqrt{\hbar D_{\mathrm{Nb}}/\Delta_{00}}$, temperature $t = T/T_c$ so $k_B T = t\,\Delta_{00}/1.7639$. The computed free energy per unit area is in units $N_0\Delta_{00}^2\xi_0$, with N_0 fixed by the condensation-energy calibration $N_0\Delta_{00}^2 = H_c(0)^2/\mu_0$ (Sec. 3) [14].

1 The Usadel equation

Literature. In the diffusive (dirty) limit the quasiclassical Green's function obeys the Usadel equation [1]. For a superconductor with an exchange field the 4×4 Nambu $\otimes$ spin, real-energy form used by Linder, Yokoyama and Sudbø [2] and (in time-dependent generality) by Simensen *et al.* [3] is, in the static limit,

$$D\,\nabla(\check{g}\,\nabla\check{g}) + i\big[\varepsilon\hat{\tau}_3 + \mathbf{h}\cdot\hat{\boldsymbol{\sigma}} + \hat{\Delta},\ \check{g}\big] = 0, \qquad \hat{\boldsymbol{\sigma}} = \mathrm{diag}(\boldsymbol{\sigma}, \boldsymbol{\sigma}^*), \tag{1}$$

where $\hat{\tau}_3 = \mathrm{diag}(1,1,-1,-1)$ and the pair-potential matrix of Ref. [3], Eq. (3), is

$$\hat{\Delta}^{(\mathrm{Simensen})} = \begin{pmatrix} 0 & \Delta\, i\sigma_y \\ \Delta^*\, i\sigma_y & 0 \end{pmatrix}. \tag{2}$$

Our form. We work in the Matsubara representation, obtained by the standard continuation $\varepsilon \to i\omega_n$, $\omega_n = \pi k_B T(2n{+}1)$ [13]. Multiplying Eq. (1) by $-i$ and collecting terms, the solver integrates

$$\boxed{\hbar D\,\partial_x(\hat{g}\,\partial_x\hat{g}) = \big[\hat{\Lambda}, \hat{g}\big], \qquad \hat{\Lambda} = \omega_n\hat{\tau}_3 + i\hat{V} + \hat{\Delta},} \tag{3}$$

$$\hat{V} = \begin{pmatrix} \mathbf{h}\cdot\boldsymbol{\sigma} & 0 \\ 0 & +(\mathbf{h}\cdot\boldsymbol{\sigma})^* \end{pmatrix}, \qquad \hat{\Delta} = \begin{pmatrix} 0 & \Delta\, i\sigma_y \\ -\Delta\, i\sigma_y & 0 \end{pmatrix} \quad (\Delta \text{ real}). \tag{4}$$

Where our form differs from the citations, and why.

1. *Sign of the lower-left block of* $\hat{\Delta}$. Our $-\Delta\, i\sigma_y$ differs from Eq. (2). This is a pure convention tied to the definition of the hole (tilde) amplitudes in the Riccati parametrisation of Sec. 2: conventions are related by a similarity transformation $\hat{g} \to S\hat{g}S^{-1}$, under which all traces (hence all observables and the free energy) are invariant. Our sign set is *fixed* by demanding that the homogeneous BCS solution takes the canonical form

$$\hat{g}_{\mathrm{bulk}} = \frac{\omega_n\hat{\tau}_3 + \hat{\Delta}}{\Omega_n}, \qquad \Omega_n = \sqrt{\omega_n^2 + \Delta^2}, \tag{5}$$

when built from the parametrisation (8) with $\gamma = \tilde{\gamma} = \frac{\Delta}{\omega_n+\Omega_n} i\sigma_y$ (verified to machine precision; test T1–T2 of the validation suite).

2. *Structure of the exchange block $\hat{V}$.* The superficially natural Bogoliubov–de Gennes guess $\mathrm{diag}(\mathbf{h}\cdot\boldsymbol{\sigma}, -(\mathbf{h}\cdot\boldsymbol{\sigma})^*)$ is *wrong in this representation*: it commutes with the singlet. For $\mathbf{h} = h\hat{z}$ the upper-right block of the commutator with $\hat{\Delta}$ is $h\Delta\,[\sigma_z(i\sigma_y) \pm (i\sigma_y)\sigma_z]$ and, using $\sigma_z(i\sigma_y) = \sigma_x = -(i\sigma_y)\sigma_z$,

$$\sigma_z(i\sigma_y) + (i\sigma_y)\sigma_z = 0 \quad \text{(no pair breaking!)}, \qquad \sigma_z(i\sigma_y) - (i\sigma_y)\sigma_z = 2\sigma_x. \tag{6}$$

With the structure (4) the collinear bulk solution reproduces the known exchange-split pair amplitudes [9, 10]

$$f_\pm(\omega_n) = \frac{\Delta}{\sqrt{(\omega_n \pm ih)^2 + \Delta^2}}, \tag{7}$$

which we verify numerically to 10^{-12} (validation T5a evaluates the free energy of this state against the exact analytic expression).

2 Riccati parametrisation

Literature. The Riccati parametrisation was introduced by Schopohl and Maki [5] and extended to spin-dependent problems with the tilde-conjugation $\tilde{\gamma}(\mathbf{r},\varepsilon) = \gamma(\mathbf{r},-\varepsilon^*)^*$ by Eschrig [6]. Diffusive, spin-full forms appear in Refs. [2, 7, 4]. We use the normalisation-preserving map (Ref. [4], Eq. (13); equivalent to Ref. [7])

$$\hat{g} = \begin{pmatrix} N(1+\gamma\tilde{\gamma}) & 2N\gamma \\ -2\tilde{N}\tilde{\gamma} & -\tilde{N}(1+\tilde{\gamma}\gamma) \end{pmatrix}, \qquad N = (1-\gamma\tilde{\gamma})^{-1}, \quad \tilde{N} = (1-\tilde{\gamma}\gamma)^{-1}, \tag{8}$$

with 2×2 matrices $\gamma(x), \tilde{\gamma}(x)$; $\hat{g}^2 = \hat{1}$ holds identically.

The cited transport equation. Equation (10) of Linder, Yokoyama and Sudbø [2], written for the *ferromagnetic* layer of an S/F structure (where the pair potential vanishes, $\Delta = 0$, and the superconductor enters only through boundary conditions), has the real-energy structure

$$D\Big[\partial_x^2\gamma + 2(\partial_x\gamma)\,\tilde{N}\tilde{\gamma}\,(\partial_x\gamma)\Big] + i\Big[2\varepsilon\,\gamma + \mathbf{h}\cdot\boldsymbol{\sigma}\,\gamma - \gamma\,(\mathbf{h}\cdot\boldsymbol{\sigma})^*\Big] = 0. \tag{9}$$

Our equations. The solver integrates, at each Matsubara frequency,

$$\hbar D\Big[\gamma'' + 2\gamma'\,\tilde{N}\tilde{\gamma}\,\gamma'\Big] = 2\omega_n\gamma + i\big(\mathbf{h}\cdot\boldsymbol{\sigma}\,\gamma - \gamma\,\mathbf{h}\cdot\boldsymbol{\sigma}^*\big) \;-\; \Delta\,i\sigma_y \;-\; \Delta\,\gamma\,(i\sigma_y)\,\gamma, \tag{10}$$

$$\hbar D\Big[\tilde{\gamma}'' + 2\tilde{\gamma}'\,N\gamma\,\tilde{\gamma}'\Big] = 2\omega_n\tilde{\gamma} + i\big(\tilde{\gamma}\,\mathbf{h}\cdot\boldsymbol{\sigma} - \mathbf{h}\cdot\boldsymbol{\sigma}^*\,\tilde{\gamma}\big) \;-\; \Delta\,i\sigma_y \;-\; \Delta\,\tilde{\gamma}\,(i\sigma_y)\,\tilde{\gamma}. \tag{11}$$

Relation to Eq. (9), term by term.

1. *Frequency term.* Continuation $\varepsilon \to i\omega_n$ turns $+2i\varepsilon\gamma$ into $-2\omega_n\gamma$; moving it to the right-hand side gives the $+2\omega_n\gamma$ of Eq. (10). The gradient and exchange structures are *identical* to Eq. (9).

2. *The pair-potential terms $-\Delta\,i\sigma_y - \Delta\,\gamma(i\sigma_y)\gamma$* — the "extra" terms you asked about — are absent from Eq. (9) *only because Ref. [2] solves the equation inside the ferromagnet where $\Delta = 0$.* Our solver treats the Nb layer self-consistently, so $\hat{\Delta}$ must appear. Two independent derivations fix their form:

   (a) Reduction to the textbook θ-parametrisation. Set $\mathbf{h} = 0$ and take the singlet ansatz $\gamma = \tilde{\gamma} = \gamma_0(x)\,i\sigma_y$. Using $(i\sigma_y)^2 = -1$ one finds $\gamma\tilde{\gamma} = -\gamma_0^2$, $\tilde{N} = (1+\gamma_0^2)^{-1}$, $\gamma'\tilde{N}\tilde{\gamma}\gamma' = -\frac{\gamma_0\gamma_0'^2}{1+\gamma_0^2}\,i\sigma_y$, and $\gamma(i\sigma_y)\gamma = -\gamma_0^2\,i\sigma_y$, so Eq. (10) collapses to the scalar equation

$$\hbar D\Big[\gamma_0'' - \frac{2\gamma_0\gamma_0'^2}{1+\gamma_0^2}\Big] = 2\omega_n\gamma_0 - \Delta\,(1-\gamma_0^2). \tag{12}$$

Substituting $\gamma_0 = \tan(\theta/2)$ and the half-angle identities $\sin\theta = 2\gamma_0/(1+\gamma_0^2)$, $\cos\theta = (1-\gamma_0^2)/(1+\gamma_0^2)$ turns Eq. (12) into

$$\frac{\hbar D}{2}\,\theta'' = \omega_n \sin\theta - \Delta\cos\theta, \tag{13}$$

the standard θ-parametrised Usadel equation of the diffusive proximity-effect literature (see, e.g., the reviews [11, 12]). Any other coefficient in front of the two Δ terms would fail this reduction.

(b) Exact elimination from the 4×4 operator. Independently of any transcription, the code's right-hand side was constructed by inserting the parametrisation (8) into the full matrix equation (3) and solving its (1,2) and (2,1) Nambu blocks — which are linear in $(\gamma'', \tilde\gamma'')$ — for the second derivatives. Equations (10)–(11) agree with this exact elimination to machine precision at random matrix arguments (validation T1). There is therefore no possibility of a sign or factor error relative to Eq. (3).

3. *Independence of $\tilde\gamma$.* In equilibrium the tilde function is defined by $\tilde\gamma(\varepsilon) = \gamma(-\varepsilon^*)^*$ [6]. For a *noncollinear* exchange texture there is no pointwise algebraic relation between $\tilde\gamma$ and γ *at the same Matsubara frequency*: we verified numerically that $\tilde\gamma \neq \gamma^*, \gamma^T, \gamma^\dagger, \sigma_y\gamma^*\sigma_y$ for conical $\mathbf{h}(x)$. Both fields are therefore propagated, giving 32 real first-order ODE components. (In the collinear bulk, $\tilde\gamma = \gamma = \frac{\Delta}{\omega_n+\Omega_n} i\sigma_y$.)

3 The free-energy functional

Literature. We evaluate the quasiclassical (Luttinger–Ward–type) free energy of Virtanen, Vargunin and Silaev [4], whose Eqs. (20)–(21) give, for the diffusive limit and with the normal state subtracted,

$$\frac{F_{sn}}{N_0} = \int dx\,\Big\{\frac{|\Delta|^2}{\lambda} - \frac{\pi T}{2}\sum_{\omega}\mathrm{tr}\Big[(\omega_n + i\,\mathbf{h}\cdot\boldsymbol{\sigma})\hat\tau_3\,\hat g + \hat\Delta\,\hat g - \frac{\hbar D}{4}\big(\partial_x\hat g\big)^2\Big]\Big\}, \tag{14}$$

whose stationary point reproduces the Usadel equation *and* the gap equation — i.e. the self-consistent solution is a saddle point of F. Reference [4] establishes this for arbitrary coordinate-dependent exchange fields, which is what licenses its use for the conical Ho texture.

Our implementation and every deviation from Eq. (14).

1. *Consistent $\hat\Lambda$.* Under a constrained variation $\delta\hat g = [\delta W, \hat g]$ (which preserves $\hat g^2 = 1$), stationarity of F requires that the linear term $\mathrm{tr}[\hat\Lambda\hat g]$ contain *the same* $\hat\Lambda$ as the commutator in Eq. (3): $\delta F \propto \mathrm{tr}\{\delta W\,(\hbar D\,\partial(\hat g\partial\hat g) - [\hat\Lambda, \hat g])\}$. We therefore use $(\omega_n\hat\tau_3 + i\hat V)$ with the $\hat V$ of Eq. (4); this coincides with the $(\omega_n + i\mathbf{h}\cdot\boldsymbol{\sigma})\hat\tau_3$ of Eq. (14) up to the similarity transformation relating the $\hat g$ conventions, under which the trace is invariant.

2. *Frequency sum.* $\sum_\omega$ runs over positive and negative ω_n; the two are complex conjugates for real Δ and real $\mathbf{h}(x)$, so we evaluate $\sum_\omega \to 2\,\mathrm{Re}\sum_{n\geq0}$, and subtract the normal state pointwise via $\hat g \to \hat g - \hat\tau_3$ (the normal state of the full structure has $\gamma = 0$, $\hat g = \hat\tau_3$ and $F \equiv 0$ exactly).

3. *Cutoff-free pairing term.* The coupling λ is eliminated with the standard BCS identity (see, e.g., [13])

$$\frac{1}{\lambda} = \ln\frac{T}{T_c} + 2\pi k_B T\sum_{n=0}^{N_c-1}\frac{1}{\omega_n}, \tag{15}$$

and the resulting Δ^2/ω_n counterterm is combined with the trace *per frequency*, so the summand decays as ω_n^{-3} in the bulk and the total is insensitive to the cutoff N_c. The

remaining slowly-decaying piece comes from the interfaces ($F_n \sim n^{-p}$, $p \simeq 1.5$–2); it is removed by fitting p on the tail and adding the analytic remainder ("tail extrapolation"; the extrapolated $dF/d\alpha$ changes by $< 0.1\%$ between $N_c = 24$ and 36, validation T8).

4. *Normalisation.* Evaluating the functional on the bulk solution (5) gives $F_{\mathrm{bulk}} = \Delta^2 \ln t + 2\pi k_B T \sum_n [\Delta^2/\omega_n - 2(\Omega_n - \omega_n)]$, which tends to $-\frac{1}{2}\Delta_{00}^2$ as $t \to 0$ and follows the BCS deviation function at finite t. Hence N_0 in our units is defined by the textbook condensation energy, $N_0 \Delta_{00}^2 = H_c(0)^2/\mu_0$ [14]; converting to SI needs only Nb's measured thermodynamic critical field.

5. *Error robustness.* Because the solution is a stationary point of F, numerical errors of size ϵ in γ or Δ perturb F only at $O(\epsilon^2)$; this is what makes sub-μJ/m^2 free-energy *differences* between magnetic configurations resolvable.

6. *Gap equation.* $\delta F/\delta\Delta = 0$ gives $\Delta(x) = 2\pi k_B T \lambda \sum_{n \geq 0} f_s(x, \omega_n)$ with $f_s = \frac{1}{4}\mathrm{tr}[\hat{P}\hat{g}]$, $\hat{P} = \partial\hat{\Delta}/\partial\Delta$; in the bulk $f_s = \Delta/\Omega_n$ and the standard gap equation is recovered.

4 Geometry, interfaces and boundary conditions

Transparent interfaces via a coordinate map. With a common normal density of states, flux conservation at a transparent interface requires $D\,\partial_x\gamma$ continuous. Defining u by $dx = (D_j/D_{\mathrm{Nb}})\,du$ makes *both* γ and $\partial_u\gamma$ continuous, so the whole stack becomes one smooth boundary-value problem in u with piecewise-constant coefficients (the equation in u acquires the factor D_j on its right-hand side). Outer surfaces carry $\partial_u\gamma = \partial_u\tilde{\gamma} = 0$ (vacuum).

Kupriyanov–Lukichev barriers. A thin layer of thickness d_b and diffusivity D_b occupies the u-length $r_b = d_b/D_b$; as $D_b \to 0$ its right-hand side vanishes, so across it $\hat{g}\,\partial_u\hat{g} = \mathrm{const}$ (conserved spectral current) while γ acquires the proportional jump

$$\Delta\gamma \;=\; r_b \left(D\,\partial_x\gamma\right), \tag{16}$$

which is the Kupriyanov–Lukichev boundary condition [8] with the dimensionless interface resistance $r_b \simeq \gamma_B$ (defined here with the Nb-side ξ_0, D_{Nb}). Because the barrier is part of the domain, the associated interface free energy is generated automatically by the gradient term of Eq. (14) integrated across it — variational consistency of F with the boundary conditions is preserved by construction. Metallic S/F interfaces correspond to $\gamma_B \sim 0.1$–2.

Magnetic texture. Py is a uniform in-plane exchange field $h_{\mathrm{Py}}\hat{z}$; Ho carries the conical texture (axis = film normal $\hat{x}$)

$$\mathbf{h}(x) = h_{\mathrm{Ho}}\left[\cos\alpha\,\hat{x} + \sin\alpha\,(\cos Qx\,\hat{y} + \sin Qx\,\hat{z})\right], \qquad Q = 2\pi/\lambda_{\mathrm{spiral}}, \tag{17}$$

as in Ref. [2], with intrinsic $\alpha_0 = 80°$ and $\lambda_{\mathrm{spiral}} = 6.3\,\mathrm{nm}$. Exchange fields of a few Δ_{00} are a *weak-ferromagnet surrogate* required for the validity of the diffusive theory (Ref. [2] makes the same substitution for Ho); trends and mechanisms are quantitative within the model, absolute magnitudes for the real materials are order-of-magnitude.

5 Implementation overview

For one magnetic configuration the solver executes:

1. **Mesh and coefficients.** Build the u-mesh (extra points to resolve the spiral); smooth all coefficient profiles over $w_{\mathrm{int}} \simeq 0.04\,\xi_0$ with tanh walls (collocation methods require smooth right-hand sides; validation T7 shows the physics is insensitive to w_{int}).

2. **Per-frequency boundary-value problems.** For each ω_n solve Eqs. (10)–(11) (32 real components: $\gamma, \tilde{\gamma}$ and first derivatives) with `scipy.solve_bvp`, warm-started from the previous outer iteration / previous configuration.

3. **Self-consistency.** Update $\Delta(x)$ from the gap equation with the cutoff-free coupling (15), accelerated by Anderson(1) mixing; detect gap collapse (normal state, $F = 0$ exactly).

4. **Final consistent pass.** Re-solve *all* frequencies at the converged Δ before evaluating F (evaluating F on stale solutions leaves a spurious $\sim 1/\omega_n$ Matsubara tail).

5. **Free energy.** Assemble Eq. (14) per frequency (traces built directly from $\gamma, \tilde{\gamma}$ via Eq. (8)), apply the tail extrapolation, and report both raw and extrapolated F.

6. **Observables.** Pair amplitudes from the (1,2) block, $f = (f_0 + \mathbf{f}_t \cdot \boldsymbol{\sigma})\, i\sigma_y$: singlet f_0, m=0 triplet $\mathbf{f}_t \cdot \hat{h}(x)$ (short-range), equal-spin $|\mathbf{f}_t - (\mathbf{f}_t \cdot \hat{h})\hat{h}|$ (long-range). Configuration sweeps (Ho cone angle α, Py angle θ_{Py}, temperature) then give torques, effective anisotropy fields [$\mu_0 H = -(\partial F/\partial\alpha)/(M_s d \sin\alpha)$; $K_{SC} = \frac{1}{2}\partial^2_\theta F/d_{\mathrm{Py}}$, $\mu_0 \Delta M_{\mathrm{eff}} = 2K_{SC}/M_s$] and their temperature dependence.

Validation suite (`tests.py`, `tests_tail.py`; all passing): T1 Riccati right-hand side $\equiv$ full 4×4 operator (machine precision); T2 bulk gap = BCS and F matches the analytic bulk functional to 5 digits, $F(t \to 0) \to -\Delta^2/2$; T3 full Usadel residual on a solved trilayer $\lesssim 10^{-3}$ of the solution scale; T4 rigid spin-rotation invariance of F to 10^{-13} (relative); T5 exact match to the analytic exchange-split bulk free energy, and monotone gap suppression in S/F bilayers; T6 Matsubara-cutoff insensitivity of the extrapolated F; T7 smoothing-width insensitivity; T8 cutoff-convergence of $dF/d\alpha$ to $< 0.1\%$.

References

[1] K. D. Usadel, Phys. Rev. Lett. **25**, 507 (1970).

[2] J. Linder, T. Yokoyama, and A. Sudbø, Phys. Rev. B **79**, 054523 (2009). [Eq. (10) therein; solved in the F region, $\Delta = 0$.]

[3] H. T. Simensen, L. G. Johnsen, J. Linder, and A. Brataas, Phys. Rev. B **103**, 024524 (2021). [Eqs. (1), (3); $\hat{\boldsymbol{\sigma}} = \mathrm{diag}(\boldsymbol{\sigma}, \boldsymbol{\sigma}^*)$ convention.]

[4] P. Virtanen, A. Vargunin, and M. Silaev, Phys. Rev. B **101**, 094507 (2020) [arXiv:1909.00992]. [Free-energy functional, Eqs. (20)–(21); Riccati map, Eq. (13).]

[5] N. Schopohl and K. Maki, Phys. Rev. B **52**, 490 (1995).

[6] M. Eschrig, Phys. Rev. B **61**, 9061 (2000).

[7] S. H. Jacobsen, J. A. Ouassou, and J. Linder, Phys. Rev. B **92**, 024510 (2015).

[8] M. Yu. Kupriyanov and V. F. Lukichev, Sov. Phys. JETP **67**, 1163 (1988).

[9] A. I. Buzdin, Rev. Mod. Phys. **77**, 935 (2005).

[10] F. S. Bergeret, A. F. Volkov, and K. B. Efetov, Rev. Mod. Phys. **77**, 1321 (2005).

[11] W. Belzig, F. K. Wilhelm, C. Bruder, G. Schön, and A. D. Zaikin, Superlattices Microstruct. **25**, 1251 (1999).

[12] A. A. Golubov, M. Yu. Kupriyanov, and E. Il'ichev, Rev. Mod. Phys. **76**, 411 (2004).

[13] N. B. Kopnin, *Theory of Nonequilibrium Superconductivity* (Oxford University Press, 2001).

[14] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. (McGraw–Hill, 1996).